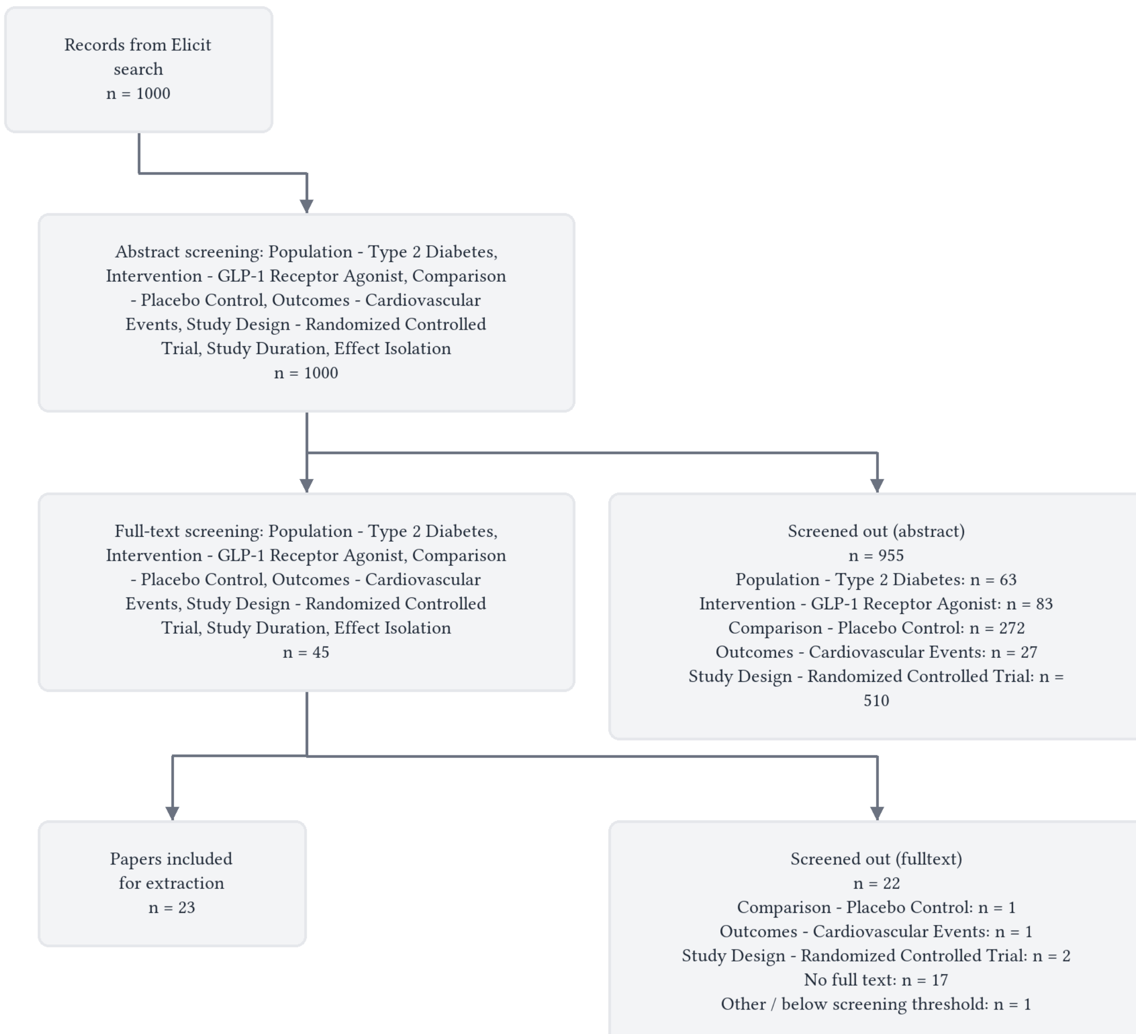
In adults with type 2 diabetes, what is the effect of GLP-1 receptor agonists compared to placebo on major adverse cardiovascular events (MACE)?

GLP-1 receptor agonists reduce major adverse cardiovascular events by approximately 13–27% compared to placebo in adults with type 2 diabetes at elevated cardiovascular risk, a benefit that is consistent across multiple agents, routes of administration, and patient subgroups.

Abstract

Five major cardiovascular outcome trials—LEADER (liraglutide, n=9,340), EXSCEL (exenatide, n=14,752), SUSTAIN 6 (subcutaneous semaglutide, n=3,297), AMPLITUDE-O (efpeglenatide, n=4,076), and SOUL (oral semaglutide, n=9,650)—evaluated GLP-1 receptor agonists versus placebo in adults with type 2 diabetes at elevated cardiovascular risk. Four of five trials demonstrated statistically superior reductions in three-point MACE (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke), with hazard ratios ranging from 0.73 (efpeglenatide) [1] to 0.87 (liraglutide) [2]. Subcutaneous semaglutide (HR 0.74; 95% CI, 0.58–0.95) [3] and oral semaglutide (HR 0.86; 95% CI, 0.77–0.96; P=0.006) [4] also showed significant benefit. EXSCEL did not achieve superiority (HR 0.91; 95% CI, 0.83–1.00; P=0.06) [5], though sensitivity analyses adjusting for unbalanced drop-in of cardioprotective medications in the placebo arm yielded a significant result (HR 0.85; P=0.008) [6]. A dose-response relationship was demonstrated with efpeglenatide (all P for trend ≤0.018) [7]. Cardiovascular benefit was consistent across subgroups defined by blood pressure [8], kidney function [9], age [10], triglyceride levels [11], ejection fraction [12], and concomitant SGLT2 inhibitor use (P-interaction=0.66) [13]. Gastrointestinal adverse events were the most common side effects across all trials [1, 2, 4], with no significant increase in pancreatitis [2, 5]. Taken together, GLP-1 receptor agonists—including both human GLP-1-based and exendin-based agents, administered subcutaneously or orally—reduce MACE by approximately 13–27% in adults with type 2 diabetes and established or high-risk cardiovascular disease.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: ”In adults with type 2 diabetes, what is the effect of GLP-1 receptor agonists compared to placebo on major adverse cardiovascular events (MACE)?”

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

Abstract screening

We screened in sources based on their abstracts that met these criteria:

- **Population - Type 2 Diabetes**: Does this study include adults (≥18 years) with confirmed type 2 diabetes mellitus as the study population (excluding studies focused on type 1 diabetes, gestational diabetes, or pediatric populations)?
- **Intervention - GLP-1 Receptor Agonist**: Does this study evaluate a GLP-1 receptor agonist (such as exenatide, liraglutide, semaglutide, dulaglutide, etc.) as monotherapy or add-on therapy?
- **Comparison - Placebo Control**: Does this study include a placebo control group (not limited to active comparators only)?
- **Outcomes - Cardiovascular Events**: Does this study report major adverse cardiovascular events (MACE) or individual cardiovascular endpoints such as cardiovascular death, myocardial infarction, stroke, or hospitalization for heart failure (not limited to only surrogate cardiovascular markers)?
- **Study Design - Randomized Controlled Trial**: Is this study a randomized controlled trial (not an observational study, case report, case series, conference abstract, or editorial)?
- **Study Duration**: Was the GLP-1 receptor agonist treatment duration at least 12 weeks?
- **Effect Isolation**: Can the effects of the GLP-1 receptor agonist be isolated in this study (i.e., is this not a fixed-dose combination product where the GLP-1 receptor agonist effects cannot be separated from other active ingredients)?

Papers that failed any strict criterion were automatically excluded. For the remaining papers, we considered all screening questions together and made a holistic judgement about whether to screen in each paper.

45 papers passed abstract screening and moved to full-text screening.

At abstract screening, the number of papers excluded for each primary reason was:

- **Population - Type 2 Diabetes**: n = 63
- **Intervention - GLP-1 Receptor Agonist**: n = 83
- **Comparison - Placebo Control**: n = 272
- **Outcomes - Cardiovascular Events**: n = 27
- **Study Design - Randomized Controlled Trial**: n = 510

Full-text screening

We then screened papers based on their full text using these additional criteria:

- **Population - Type 2 Diabetes**: Does this study include adults (≥18 years) with confirmed type 2 diabetes mellitus as the study population (excluding studies focused on type 1 diabetes, gestational diabetes, or pediatric populations)?
- **Intervention - GLP-1 Receptor Agonist**: Does this study evaluate a GLP-1 receptor agonist (such as exenatide, liraglutide, semaglutide, dulaglutide, etc.) as monotherapy or add-on therapy?
- **Comparison - Placebo Control**: Does this study include a placebo control group (not limited to active comparators only)?

- **Outcomes - Cardiovascular Events**: Does this study report major adverse cardiovascular events (MACE) or individual cardiovascular endpoints such as cardiovascular death, myocardial infarction, stroke, or hospitalization for heart failure (not limited to only surrogate cardiovascular markers)?
- **Study Design - Randomized Controlled Trial**: Is this study a randomized controlled trial (not an observational study, case report, case series, conference abstract, or editorial)?
- **Study Duration**: Was the GLP-1 receptor agonist treatment duration at least 12 weeks?
- **Effect Isolation**: Can the effects of the GLP-1 receptor agonist be isolated in this study (i.e., is this not a fixed-dose combination product where the GLP-1 receptor agonist effects cannot be separated from other active ingredients)?

Papers that failed any strict criterion were automatically excluded. For the remaining papers, we considered all screening questions together and made a holistic judgement about whether to include each paper in the final analysis.

23 papers passed full-text screening and moved to data extraction.

At full-text screening, the number of papers excluded for each primary reason was:

- **Comparison - Placebo Control**: n = 1
- **Outcomes - Cardiovascular Events**: n = 1
- **Study Design - Randomized Controlled Trial**: n = 2
- **No full text**: n = 17
- **Other / below screening threshold**: n = 1

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design**:

  Extract study design characteristics relevant to assessing cardiovascular outcomes in type 2 diabetes, including:

- Study type (RCT, open-label, etc.)
- Randomization method and blinding
- Total sample size
- Median follow-up duration
- Number of study sites and countries
- Primary vs secondary analysis (specify if MACE was primary or secondary endpoint)

- **GLP-1 RA Intervention**:

  Extract details about the GLP-1 receptor agonist intervention for cardiovascular outcomes in type 2 diabetes, including:

- Specific GLP-1 RA name (liraglutide, exenatide, efpeglenatide, etc.)
- Dose and dosing regimen
- Route of administration
- Duration of treatment
- Comparison to placebo or other control
- Concomitant standard care/background therapy

- **Population Characteristics**:

  Extract baseline characteristics of adults with type 2 diabetes relevant to cardiovascular risk, including:

- Mean age and percentage male
- Diabetes duration (mean years)
- Baseline HbA1c
- Percentage with established cardiovascular disease
- Percentage with specific CV risk factors (hypertension, dyslipidemia, smoking)
- Baseline renal function if reported
- Inclusion/exclusion criteria related to cardiovascular risk

- **MACE Definition**:

  Extract the specific definition of major adverse cardiovascular events (MACE) used in each study, including:

- Exact components included in primary MACE composite (CV death, nonfatal MI, nonfatal stroke, etc.)
- Definition of expanded MACE if reported
- Time-to-first-event vs recurrent events analysis
- Any variations from standard 3-point MACE definition

- **MACE Results**:

  Extract results for major adverse cardiovascular events comparing GLP-1 receptor agonists to placebo in type 2 diabetes, including:

- Primary MACE composite: number of events, event rates, hazard ratio with 95% CI, p-value
- Individual MACE components: CV death, nonfatal MI, nonfatal stroke (events, HR, 95% CI)
- Time-to-event analysis results
- Whether superiority or non-inferiority was demonstrated
- Expanded MACE results if reported

- **Other CV Outcomes**:

  Extract results for other cardiovascular outcomes beyond primary MACE in type 2 diabetes studies, including:

- All-cause mortality (events, HR, 95% CI)
- Heart failure hospitalization
- Coronary revascularization
- Hospitalization for unstable angina
- Any other CV endpoints reported
- Specify if these were prespecified secondary endpoints

- **Safety Outcomes**:

  Extract safety and adverse event data for GLP-1 receptor agonists vs placebo in adults with type 2 diabetes, including:

- Most common adverse events leading to discontinuation
- Gastrointestinal adverse events (nausea, vomiting, diarrhea)
- Pancreatitis incidence
- Serious adverse events
- Any safety signals of special interest

- Overall discontinuation rates
- **Statistical Analysis**:

  Extract statistical analysis methods used for cardiovascular outcomes in type 2 diabetes studies, including:

- Primary analysis approach (intention-to-treat, per-protocol)
- Time-to-event analysis methods
- Handling of competing risks
- Multiple testing adjustments
- Non-inferiority margins if applicable
- Cox proportional hazards assumptions

Results

Characteristics of Included Studies

The 23 sources encompass five major cardiovascular outcome trials (CVOTs) of GLP-1 receptor agonists—LEADER (liraglutide), EXSCEL (exenatide), SUSTAIN 6/PIONEER 6 (semaglutide), AMPLITUDE-O (efpeglenatide), and SOUL (oral semaglutide)—along with multiple prespecified secondary analyses, post hoc analyses, and one trial design paper. All primary trials were randomized, double-blind, placebo-controlled, and used three-point MACE (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke) as their primary endpoint. The table below summarizes the characteristics of all included sources.

Study	Full text retrieved?	Analysis Type	Trial	GLP-1 RA	Dose / Route	Sample Size	Median Follow-up	Population CV Risk
H. Gerstein et al., 2021	Yes	Primary CVOT [1]	AMPLITUDE-O	Efpeglenatide [1]	4 or 6 mg weekly SC [1]	4,076 [1]	1.81 years [1]	89.6% established CVD [1]
S. Marso et al., 2016	Yes	Primary CVOT [2]	LEADER	Liraglutide [2]	1.8 mg daily SC [2]	9,340 [2]	3.8 years [2]	81.3% established CVD [2]
R. Holman et al., 2017	Yes	Primary CVOT [5]	EXSCEL	Exenatide [5]	2 mg weekly SC [5]	14,752 [5]	3.2 years [5]	73.1% established CVD [5]
M. Husain et al., 2020	Yes	Pooled post hoc [3]	SUSTAIN 6 + PIONEER 6	Semaglutide [3]	0.5–1.0 mg weekly SC; 14 mg daily oral [3]	6,480 [3]	2.1 yr / 1.3 yr [3]	Prior MI 34.3%, prior stroke/TIA 15.8% [3]

Study	Full text retrieved?	Analysis Type	Trial	GLP-1 RA	Dose / Route	Sample Size	Median Follow-up	Population CV Risk
S. Verma et al., 2019	Yes	Post hoc (LEADER) [14]	LEADER	Liraglutide [14]	Up to 1.8 mg daily [14]	9,340 [14]	3.8 years [14]	Age ≥50 with CVD or ≥60 with risk factors [14]
S. Verma et al., 2018	Yes	Post hoc (LEADER) [15]	LEADER	Liraglutide [15]	1.8 mg or max tolerated [15]	9,340 [15]	3.8 years [15]	Polyvascular disease sub-groups [15]
J. Buse et al., 2020	Yes	Mediation analysis (LEADER) [16]	LEADER	Liraglutide [16]	1.8 mg daily [16]	9,340 [16]	3.8 years [16]	HbA1c ≥7%, high CV risk [16]
J. Mann et al., 2017	Yes	Prespecified sec-ondary (LEADER) [17]	LEADER	Liraglutide [17]	Not specified [17]	9,340 [17]	3.84 years [17]	Mean eGFR 80 mL/min/1.73 m² [17]
R. Mentz et al., 2018	Yes	Prespecified subgroup (EXSCEL) [18]	EXSCEL	Exenatide [18]	2 mg weekly [18]	14,752 [18]	Not specified [18]	~70% with known CVD [18]
L.A. Leiter et al., 2020	Yes	Post hoc (LEADER + SUSTAIN 6) [8]	LEADER + SUSTAIN 6	Liraglutide / Semaglu-tide [8]	SC injection [8]	9,340 / 3,297 [8]	3.8 yr / 2.1 yr [8]	Stage 1 HTN ~30%, Stage 2 ~41–43% [8]
J. Mann et al., 2018	Yes	Post hoc (LEADER) [19]	LEADER	Liraglutide [19]	1.8 mg daily [19]	9,340 (implied) [19]	Not specified [19]	eGFR <60: mean 45.7; eGFR ≥60: mean 90.8 [19]
MD N. Marx et al., 2025	Yes	Prespecified (SOUL) [13]	SOUL	Oral semaglu-tide [13]	3⟶7⟶14 mg daily oral [13]	9,650 [13]	47.5 months [13]	ASCVD or CKD; mean age 65–67 yr [13]
J.S. Neves et al., 2024	Yes	Post hoc (EXSCEL) [12]	EXSCEL	Exenatide [12]	Once weekly [12]	14,752 [12]	Not specified [12]	Stratified by LVEF [12]

Study	Full text retrieved?	Analysis Type	Trial	GLP-1 RA	Dose / Route	Sample Size	Median Follow-up	Population CV Risk
M. Husain et al., 2020a	Yes	Combined post hoc (SUSTAIN + PIONEER) [20]	SUSTAIN + PIONEER	Semaglutide [20]	0.5–1.0 mg SC; 3–14 mg oral [20]	Large pooled cohort [20]	Up to 2 years [20]	Mean age 59.9 yr; eGFR 88.7 [20]
P. Rossing et al., 2023	Yes	Post hoc (SUSTAIN 6 + PIONEER 6) [9]	SUSTAIN 6 + PIONEER 6	Semaglutide [9]	0.5–1.0 mg SC; 14 mg oral [9]	6,461 [9]	2.1 yr / 15.9 mo [9]	Stratified by eGFR and UACR [9]
M. Gilbert et al., 2018	Yes	Post hoc (LEADER) [10]	LEADER	Liraglutide [10]	Not specified [10]	9,340 [10]	Not specified [10]	Stratified by age; ~17 yr diabetes duration in ≥75 yr group [10]
S. Verma et al., 2023	Yes	Post hoc (SUSTAIN 6 + PIONEER 6) [11]	SUSTAIN 6 + PIONEER 6	Semaglutide [11]	0.5–1.0 mg SC; 14 mg oral [11]	6,417 [11]	2 yr / 15.9 mo [11]	Mean age 65.4 yr; mean eGFR 75.0 [11]
A. Bethel et al., 2020	Yes	Sensitivity analysis (EXSCEL) [6]	EXSCEL	Exenatide [6]	2 mg weekly [6]	14,752 [6]	3.2 years [6]	~70% with previous CVD [6]
R. Holman et al., 2016	Yes	Trial design paper [21]	EXSCEL	Exenatide [21]	2 mg weekly SC [21]	~14,000 [21]	Expected >3 yr [21]	~70% with established CVD [21]
S. Verma et al., 2020	Yes	Post hoc (LEADER + SUSTAIN 6) [22]	LEADER + SUSTAIN 6	Liraglutide / Semaglutide [22]	Liraglutide 1.8 mg; semaglutide 0.5–1.0 mg [22]	9,340 / 3,297 [22]	Not specified [22]	Microvascular disease subgroups [22]
H. Gerstein et al., 2023	Yes	Dose-response (AMPLITUDE-O) [7]	AMPLITUDE-O	Efpeglenatide [7]	4 mg and 6 mg weekly SC [7]	4,076 [7]	1.8 years [7]	89.5% established CVD [7]

Study	Full text retrieved?	Analysis Type	Trial	GLP-1 RA	Dose / Route	Sample Size	Median Follow-up	Population CV Risk
K. Gooding et al., 2024	Yes	Post hoc (EXSCEL) [23]	EXSCEL	Exenatide [23]	2 mg weekly SC [23]	14,752 [23]	3.2 years [23]	73.1% established CVD; median diabetes 12 yr [23]
Unveiling CV Benefits, 2025	No	Narrative review of SOUL [4]	SOUL	Oral semaglutide [4]	3⟶7⟶14 mg daily oral [4]	9,650 [4]	49.5 months [4]	75% established CVD; mean age 66 yr [4]

Across the five CVOTs, sample sizes ranged from 4,076 (AMPLITUDE-O) [1] to 14,752 (EXSCEL) [5], and median follow-up ranged from 1.81 years [1] to approximately 4 years (LEADER [2] and SOUL [4]). All trials enrolled high-cardiovascular-risk populations, though the proportion with established cardiovascular disease varied from approximately 73% in EXSCEL [5] to nearly 90% in AMPLITUDE-O [1]. The GLP-1 RA agents included human GLP-1-based molecules (liraglutide, semaglutide) and exendin-based molecules (exenatide, efpeglenatide), administered via subcutaneous injection or orally (semaglutide only) [4, 13]. All trials used standard three-point MACE as the primary composite endpoint [1, 2, 4, 5, 13], employed intention-to-treat analyses [1, 2, 4, 5, 13], and applied Cox proportional hazards models for time-to-first-event analysis.

Effects on MACE

Primary MACE Outcomes Across Trials

Trial	GLP-1 RA	Events (Drug vs. Placebo)	HR (95% CI)	Noninferiority	Superiority
LEADER	Liraglutide	608 (13.0%) vs. 694 (14.9%) [2]	0.87 (0.78–0.97) [2]	P<0.001 [2]	P=0.01 [2]
EXSCEL	Exenatide	839 (11.4%) vs. 905 (12.2%) [5]	0.91 (0.83–1.00) [5]	P<0.001 [5]	P=0.06 [5]
SUSTAIN 6	SC semaglutide	—	0.74 (0.58–0.95) [3]	—	Demonstrated [3]
PIONEER 6	Oral semaglutide	—	0.79 (0.57–1.11) [3]	—	Not demonstrated [3]
AMPLITUDE-O	Efpeglenatide (combined)	189 (7.0%) vs. 125 (9.2%) [1]	0.73 (0.58–0.92) [1]	P<0.001 [1]	P=0.007 [1]
SOUL	Oral semaglutide	579 (12.0%) vs. 668 (13.8%) [4]	0.86 (0.77–0.96) [4]	Met [4]	P=0.006 [4]

Four of the five CVOTs demonstrated both noninferiority and superiority for MACE reduction. LEADER showed a

13% relative risk reduction with liraglutide (HR 0.87; 95% CI, 0.78–0.97; P=0.01 for superiority) [2]. AMPLITUDE-O demonstrated the largest point estimate of benefit with efpeglenatide (HR 0.73; 95% CI, 0.58–0.92; P=0.007) [1], though this trial had the shortest follow-up (1.81 years) [1] and highest proportion of patients with established CVD (89.6%) [1]. SUSTAIN 6 demonstrated superiority for subcutaneous semaglutide (HR 0.74; 95% CI, 0.58–0.95) [3], and the SOUL trial confirmed superiority for oral semaglutide over a longer follow-up of approximately 4 years (HR 0.86; 95% CI, 0.77–0.96; P=0.006) [4]. EXSCEL was the sole trial that met noninferiority but failed to demonstrate superiority for once-weekly exenatide (HR 0.91; 95% CI, 0.83–1.00; P=0.06) [5].

Individual MACE Components

Data on individual MACE components were reported in varying detail. In LEADER, cardiovascular death was significantly reduced (HR 0.78; 95% CI, 0.66–0.93; P=0.007), while nonfatal MI and nonfatal stroke showed nonsignificant trends toward benefit [2]. In EXSCEL, none of the individual components differed significantly between groups [5]. In the AMPLITUDE-O dose-response analysis, the 6 mg efpeglenatide dose reduced cardiovascular death (HR 0.55; 95% CI, 0.35–0.88; P=0.012) and nonfatal MI (HR 0.64; 95% CI, 0.43–0.96; P=0.033), but not nonfatal stroke (HR 0.79; 95% CI, 0.47–1.34; P=0.38) [7]. Combined semaglutide data showed a significant reduction in fatal and nonfatal stroke (HR 0.68; 95% CI, 0.46–1.00), while MI was not significantly reduced (HR 0.89; 95% CI, 0.67–1.18) [3].

All-Cause Mortality and Other Cardiovascular Outcomes

Trial	All-Cause Mortality HR (95% CI)	HF Hospitalization HR (95% CI)	Expanded MACE HR (95% CI)
LEADER	0.85 (0.74–0.97) [2]	0.87 (0.73–1.05) [2]	Not specified [2]
EXSCEL	0.86 (0.77–0.97), nominal P=0.016 [6]	0.94 (0.78–1.13) [23]	Not significant [5]
AMPLITUDE-O	0.78 (0.58–1.06) [1]	0.61 (0.38–0.98) [1]	0.79 (0.65–0.96) [1]
SUSTAIN 6 + PIONEER 6	Not reported [3]	1.03 (0.75–1.40) [3]	Not reported [3]
SOUL	Not reported [4]	0.82, P=0.013 [4]	Not reported [4]

Liraglutide significantly reduced all-cause mortality by 15% (HR 0.85; 95% CI, 0.74–0.97) [2], while exenatide showed a nominally significant 14% reduction (HR 0.86; P=0.016) [6]. For heart failure hospitalization, results were generally neutral across semaglutide trials (HR 1.03) [3] and EXSCEL (HR 0.94) [23], but significant reductions were observed in AMPLITUDE-O (HR 0.61; 95% CI, 0.38–0.98) [1] and SOUL (HR 0.82; P=0.013) [4].

Dose-Response Relationship

The AMPLITUDE-O dose-response analysis revealed a graded relationship between efpeglenatide dose and cardiovascular outcomes. The 6 mg dose showed a significant 35% MACE reduction (HR 0.65; 95% CI, 0.50–0.86; P=0.003), while the 4 mg dose showed a nonsignificant 18% reduction (HR 0.82; 95% CI, 0.63–1.06; P=0.14) [7]. A clear dose-response trend was observed across all primary and secondary outcomes (all P for trend ≤0.018) [7]. This dose-response pattern extended to kidney outcomes: the composite renal endpoint showed HR 0.63 for 6 mg (P<0.0001) and HR 0.73 for 4 mg (P=0.0009) [7].

Renal Outcomes

Renal outcomes were reported as secondary endpoints in several trials. In LEADER, the composite renal outcome (new-onset persistent macroalbuminuria, persistent doubling of serum creatinine, end-stage renal disease, or renal death) was significantly reduced with liraglutide (HR 0.78; 95% CI, 0.67–0.92; P=0.003), driven primarily by reduced new-onset macroalbuminuria (HR 0.74; 95% CI, 0.60–0.91; P=0.004) [17]. AMPLITUDE-O similarly showed a significant reduction in the composite renal outcome (HR 0.68; 95% CI, 0.57–0.79; P<0.001) [1].

Mediation Analysis

An exploratory mediation analysis from LEADER identified HbA1c (up to 41–83% mediation depending on analytical method) and urinary albumin-to-creatinine ratio (up to 29–33% mediation) as potential mediators of liraglutide's cardiovascular benefit [16]. Mediation effects were small for body weight, systolic blood pressure, LDL cholesterol, confirmed hypoglycemia, sulfonylurea use, and insulin use [16].

Subgroup Analyses

Multiple post hoc and prespecified subgroup analyses examined whether the cardiovascular benefits of GLP-1 RAs were consistent across clinically relevant subpopulations.

Subgroup	Trial(s)	Key Findings
Baseline blood pressure categories	LEADER, SUSTAIN 6	No heterogeneity across BP categories for MACE (P-interaction = 0.06 and 0.40) [8]
CKD (eGFR <60 vs. ≥60)	LEADER	Greater MACE reduction in eGFR <60 (HR 0.69; 95% CI, 0.57–0.85) vs. eGFR ≥60 (HR 0.94; 95% CI, 0.83–1.07; P-interaction=0.01) [19]
Kidney function (eGFR/UACR)	SUSTAIN 6 + PIONEER 6	Consistent MACE reduction across all eGFR and UACR subgroups (P-interaction >0.05) [9]
Polyvascular disease	LEADER	HR 0.82 (95% CI, 0.66–1.02) for polyvascular; HR 0.82 (95% CI, 0.71–0.95) for single vascular [15]
Age (≥75 vs. 60–74 yr)	LEADER	34% MACE reduction in ≥75 yr vs. lesser reduction in 60–74 yr (P-interaction=0.054) [10]
Microvascular disease	LEADER, SUSTAIN 6	Patients with microvascular disease had higher baseline MACE risk (HR 1.15 in LEADER, 1.56 in SUSTAIN 6) [22]; benefit of GLP-1 RA consistent regardless of microvascular disease status [22]
Baseline triglycerides	SUSTAIN 6 + PIONEER 6	Consistent MACE reduction across TG levels [11]

Subgroup	Trial(s)	Key Findings
CV risk continuum	SUSTAIN + PIONEER pooled	Reduced relative and absolute risk across entire CV risk continuum; relative risk reduction tended to be largest with low CV risk score, while absolute risk reduction was greatest at intermediate-to-high CV risk [20]
LVEF (<40% vs. ≥40%)	EXSCEL	MACE HR 0.96 (0.70–1.31) for LVEF <40% vs. 0.84 (0.71–0.98) for LVEF ≥40% (P-interaction=0.47) [12]
Concomitant SGLT2i use	SOUL	HR 0.89 (0.71–1.11) with SGLT2i vs. 0.84 (0.74–0.95) without (P-interaction=0.66) [13]
Concomitant SU use	EXSCEL	No modification by SU use or SU type (P-interaction=0.88 and 0.78) [23]
First vs. recurrent MACE	LEADER	15.7% reduction in total (first + recurrent) MACE (HR 0.84; 95% CI, 0.76–0.93) [14]

The cardiovascular benefits of GLP-1 RAs appear broadly consistent across most subgroups examined. An apparent exception was the LEADER post hoc analysis stratifying by baseline eGFR, which showed a significantly greater MACE reduction in patients with eGFR <60 (HR 0.69) than those with eGFR ≥60 (HR 0.94; P-interaction=0.01) [19]. However, this interaction was not confirmed when eGFR was analyzed as a continuous variable (P-interaction=0.61) or across finer eGFR subgroups (P-interaction=0.13) [19], and pooled semaglutide data from SUSTAIN 6 and PIONEER 6 showed consistent benefit across all kidney function categories (P-interaction >0.05) [9]. Concomitant use of SGLT2 inhibitors did not attenuate the MACE benefit of oral semaglutide in the SOUL trial (P-interaction=0.66) [13], suggesting additive cardiovascular protection.

Safety Outcomes

Trial	GI Adverse Events	Pancreatitis	Serious AEs	Discontinuation Rate
LEADER	Nausea, vomiting, diarrhea more common with liraglutide [2]	18 vs. 23 (NS) [2]	More common with liraglutide [2]	Higher with liraglutide [2]
EXSCEL	Not detailed [5]	No significant difference [5]	No significant difference [5]	Up to 45% overall [5]
AMPLITUDE-O	Nausea, vomiting, diarrhea, bloating more frequent with efpeglenatide [1]	Slightly higher, NS [1]	Severe GI events more common [1]	5.4% vs. 3.6% [1]

Trial	GI Adverse Events	Pancreatitis	Serious AEs	Discontinuation Rate
SOUL	GI AEs: 16.6% vs. 8.2% [4]	No significant increase [4]	Similar between groups [4]	Adherence: 82.5% vs. 87.7% [4]
LEADER (age ≥75)	GI disorders 46.6% vs. 33.0% [10]	Not reported [10]	63.5% in ≥75 yr group [10]	Not reported [10]

Gastrointestinal adverse events were consistently the most common side effects across all GLP-1 RA trials and the most frequent reason for treatment discontinuation [1, 2, 4]. In the SOUL trial, gastrointestinal events occurred in 16.6% of semaglutide-treated patients versus 8.2% with placebo [4]. Pancreatitis rates were numerically similar or nonsignificantly different between treatment groups across all trials [1, 2, 4, 5]. In LEADER, gallstone disease was more common with liraglutide [2]. The safety profile of oral semaglutide was not altered by concomitant SGLT2 inhibitor use [13]. In the EXSCEL trial, no safety signal was detected based on left ventricular ejection fraction, though serious adverse events were numerically more frequent in participants with LVEF <40% receiving exenatide (25.3% vs. 18.5% with placebo) [12]. The high overall discontinuation rate of up to 45% in EXSCEL [5] is notable and may have reduced the power to detect a superiority signal.

Sensitivity Analyses and Methodological Considerations

The EXSCEL trial warrants particular attention given its failure to achieve statistical superiority. A key methodological consideration was the unbalanced drop-in of open-label glucose-lowering medications, with 38.1% of placebo patients versus 28.8% of exenatide patients receiving additional therapies during follow-up, including SGLT2 inhibitors (10.3% vs. 8.1%) and open-label GLP-1 RAs (3.4% vs. 2.4%) [6]. When inverse probability of treatment weighting was applied to account for this imbalance, the MACE hazard ratio decreased from 0.91 (P=0.061) to 0.85 (P=0.008), and the all-cause mortality HR decreased from 0.86 (P=0.016) to 0.81 (P=0.012) [6]. This analysis suggests that the observed MACE result in EXSCEL may have been attenuated by greater use of cardioprotective drop-in medications in the placebo group, and that the true treatment effect of exenatide may be larger than the primary analysis indicated.

Synthesis

The overall evidence from five major CVOTs consistently supports that GLP-1 receptor agonists reduce MACE in adults with type 2 diabetes at elevated cardiovascular risk, though the magnitude and statistical significance of this benefit varied across trials. Four of five trials demonstrated statistical superiority for the primary MACE endpoint, with hazard ratios ranging from 0.73 (efpeglenatide, AMPLITUDE-O) [1] to 0.87 (liraglutide, LEADER) [2]. The SOUL trial, reporting most recently with a longer follow-up of nearly 4 years, demonstrated a 14% MACE reduction with oral semaglutide (HR 0.86; P=0.006) [4], establishing efficacy via the oral route.

The apparent outlier—EXSCEL's failure to reach statistical superiority (HR 0.91; P=0.06) [5] —can be substantially explained by trial-specific factors rather than a true lack of efficacy for exenatide. First, EXSCEL enrolled a broader population with a lower baseline cardiovascular risk (73.1% with established CVD [5] vs. 81–90% in other trials [1, 2]), diluting the population most likely to benefit. Second, the high discontinuation rate (up to 45%) [5] reduced on-treatment exposure. Third, and perhaps most importantly, the greater drop-in of open-label cardioprotective glucose-lowering medications in the placebo arm attenuated the between-group difference; IPTW correction produced a MACE HR of 0.85 (P=0.008) [6], fully consistent with the other trials. In the prespecified subgroup with established cardiovascular disease, exenatide achieved statistical significance for MACE reduction (P=0.047) [18].

The consistency of benefit across subgroups is notable. GLP-1 RA cardiovascular benefit was maintained regardless of baseline blood pressure category [8], triglyceride levels [11], microvascular disease status [22], age [10], and ejection fraction [12]. The benefit persisted when SGLT2 inhibitors were used concomitantly (P-interaction=0.66) [13], a finding of increasing clinical relevance as dual therapy becomes more common. The suggestion of enhanced benefit in patients with CKD in one LEADER analysis (P-interaction=0.01 for eGFR <60 vs. ≥60) [19] was not confirmed as a continuous interaction [19] and was not replicated in semaglutide trials [9], making it more likely a chance subgroup finding, though patients with CKD clearly derive at minimum comparable benefit.

The dose-response relationship demonstrated with efpeglenatide (P for trend ≤0.018 across all outcomes) [7] has potential class-wide implications, suggesting that titration to higher doses may optimize cardiovascular and renal benefit. Mechanistically, the mediation analysis from LEADER points to glycemic control (HbA1c mediating up to 41–83% of the effect) and possibly albuminuria reduction (29–33%) as key pathways, while weight loss, blood pressure, and lipid changes appear to contribute minimally [16]. This aligns with the concordant renal benefits observed across liraglutide (HR 0.78 for composite renal outcome) [17] and efpeglenatide (HR 0.68) [1], suggesting a shared cardiorenal mechanism.

In aggregate, the evidence supports a cardiovascular class effect for GLP-1 receptor agonists in patients with type 2 diabetes and established or high-risk cardiovascular disease, with both human GLP-1-based (liraglutide, semaglutide) and exendin-based (exenatide, efpeglenatide) agents demonstrating benefit. The magnitude of MACE reduction across the class is approximately 13–27%, driven primarily by reductions in cardiovascular death and myocardial infarction rather than stroke, with consistent safety profiles characterized predominantly by gastrointestinal adverse events.

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